Point Cloud to Map Registration Under Rigid Transformations: Cramer-Rao Bound

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***Abstract*—The common practice for evaluating the performance of point cloud registration algorithms is to compare them on publicly available registration benchmarks. However, no analysis of the algorithmic solutions relative to a universal bound is available. In this paper we provide a simple derivation of a closed form, easy to compute expression for the Cramer-Rao lower bound on the error covariance of any unbiased estimator of the rigid transformation parameters in the problem of point cloud to known landmarks registration. This type of formulation is encountered for example in problems of self-localization and navigation relative to a map or relative to known markers such as in cell-tower based navigation, as well as in medical imaging where pre-installed markers are employed. The derivation assumes that all putative matches between the observed point cloud and an *a-priori known* reference point cloud are correct, *i.e.*, all matches are *inliers*. We further compare the performance the maximum-likelihood and constrained LS algorithms with the theoretical lower bound. Furthermore, while existing analyses of the performance in estimating rigid transformations are restricted to naive models of additive isotropic measurement noise, we derive the bound subject to more general noise models, aiming at modeling both the sampling discrepancies of the point clouds, as well as the point-wise geometry-dependent statistics of the noise. We finally derive a CRB-based selection rule for rigorously establishing the "inlier threshold" which defines which putative matches are considered inliers - thus providing a principled alternative to currently employed heuristics.**

***Index Terms*—Point Clouds, Rigid Transformation Estimation, Cramer-Rao Bound**

I. Introduction

Registration of point clouds is defined as the alignment of point cloud observations on a 3D object, or scene, into a single coordinate system. A point cloud $\mathcal{P}$ is a set of points in $\mathbb{R}^3$. In many applications, these points are samples from a physical object, $\mathcal{O} \subset \mathbb{R}^3$. Viewing point clouds as sets of samples, the registration problem may be formulated as follows: Let $\mathcal{O} \subset \mathbb{R}^3$ be a physical object and $T : \mathbb{R}^3 \to \mathbb{R}^3$, a rigid transformation (*i.e.*, $T \in SE(3)$, is an element in the Special Euclidean Group). The operation of T on the object $\mathcal{O}$ is defined by $T(\mathcal{O}) := \{T(\mathbf{x}) \ : \ \mathbf{x} \in \mathcal{O}\}$. Let $\mathcal{P}$ and $\mathcal{Q}$ be two point clouds sampled from the object (or scene) $\mathcal{O}$ and the transformed object (scene) $T(\mathcal{O})$, respectively, *i.e.*, $\mathcal{P} \subset \mathcal{O}$ and $\mathcal{Q} \subset T(\mathcal{O})$. In this paper we consider a special case of the general problem, namely, the problem of point cloud to a known map registration. This type of formulation is encountered for example in problems of self-localization and navigation relative to a map or relative to known triangulation markers, as well as in medical imaging where pre-installed markers are employed.

The most common approach for point cloud registration, still employed by modern, data-driven methods, is to estimate a set of matching points, followed by outlier rejection using RANSAC [1] and constrained least squares estimation *e.g.* [2], [3]. Since the estimated point correspondences may contain outliers and sampling noise, this estimation is considered as an approximate alignment. Thus, an additional step is usually performed, where high accuracy is required, refining the initial estimate using local optimization (*e.g.* [4]–[7].) Algorithms that follow this approach, where the emphasis is on the feature extraction stage, include [8]–[15]. On the other hand [16]–[19] concentrate on the outlier rejection problem while relying on the same initial matching. Nevertheless, solutions like [14], [20] propose both a feature-extracting function and an outlier rejection mechanism.

While different solutions to the point cloud registration problem have been proposed, the common practice for evaluating their performance (absolute and relative) is to compare their performance on publicly available data sets. However, no analysis of the algorithmic solutions relative to a universal bound is available, to the best of our knowledge. Hence, in this paper, we provide a simple derivation of a closed form, easy to compute, expression for the Cramer-Rao lower bound on the error covariance of any unbiased estimator of the rigid transformation parameters in the above scenario.

The framework of the bound evaluated in this paper is applicable for example to estimation using RANSAC when identification of a set of inliers is successful, or in using ICP in the case where it successfully converges. Since most registration algorithms, old and new, rely on RANSAC for initial alignment and on ICP for refined final alignment (*e.g.* [10], [12], [21]), while most existing implementations of both RANSAC and ICP employ the constrained LS, [2], as their transformation estimation backbone, analyzing the CRB in this setup is widely applicable and is valid for almost any existing registration algorithm.

II. The Transformation Model

In the problem we address in this paper, a physical object undergoes a rigid transformation and our goal is to estimate the transformation relating any two observations on that object. The observations are in the form of point clouds obtained by some 3-D scanning device. Due to the interplay between the scanning mechanism and the 3-D shape of the moving object, the ideal, noise-free, rigid transformation model relating a point $\mathbf{p}_k$ in $\mathcal{P}$, with its corresponding point $\mathbf{q}_k$ in $\mathcal{Q}$, given by

$$\mathbf{q}_k = \mathbf{R}\mathbf{p}_k + \mathbf{t} \quad \mathbf{q}_k \in \mathcal{Q}, \quad \mathbf{p}_k \in \mathcal{P} \tag{1}$$

where $\mathbf{R}$ denotes the rotation matrix and $\mathbf{t}$ is the translation vector, *does not hold in practice*. More specifically, we consider here a model identified in applications where *a-priori* inserted markers are employed as an aid for estimating the rigid transformation of an object, such as in some medical applications. In this case, the plan is to place the markers at **known** locations $\mathcal{P}$. However, due to errors and noise in the placement procedure the markers are, in fact, placed at $\mathbf{y}_k = \mathbf{p}_k + \mathbf{n}_k$. Thus, $\mathbf{y}_k \sim N(\mathbf{p}_k, \mathbf{\Lambda}_k)$ where $\mathbf{\Lambda}_k$ models the covariance of the sampling errors at the k-th point, which can be specific to each point depending on the geometry of the object. Next, the object with the markers installed undergoes a rigid transformation which should be estimated by detecting the markers, followed by estimating the transformation based on the detected positions in the observation, and their known planned positions. We note that in the problem set-up considered in the following it is assumed that the point matching problem has already been solved, *i.e.*, for every point $\mathbf{p}_k \in \mathcal{P}$, its corresponding point in $\mathcal{Q}$ is known, even if the observations on both are noisy. In other words, it is assumed that the correspondence matching problem is correctly addressed and the assumed matches are all "inliers".

In the following, the errors in the sampling patterns of the point clouds are modeled by a multivariate Gaussian model such that the perturbed points approximately preserve the unique geometry of the object surface at each point. It is assumed that the noise at any point of the observed point cloud is independent of the noise at any other point. Thus, in this problem, the observation equation for the k-th marker is given by

$$\begin{aligned} \mathbf{z}_k &= \mathbf{R}\mathbf{y}_k + \mathbf{t} + \mathbf{w}_k \ , \quad k = 1, \ldots, N \\ &= \mathbf{R}(\mathbf{p}_k + \mathbf{n}_k) + \mathbf{t} + \mathbf{w}_k \end{aligned} \tag{2}$$

where we assume that the observation noise $\mathbf{w}_k \sim N(\mathbf{0}, \mathbf{\Xi}_k)$ where $\mathbf{\Xi}_k$ models the covariance of the measurement noise in acquiring $\mathcal{Q}$, which can be specific to each point depending on the geometry and scanning method. Thus, while it is a common practice in the literature to model the measurement noise as isotropic and white, in fact as noted above, it is a result of the depth measurement noise at every point and its projection on the coordinate axes. Hence, we have that

$$\mathbf{z}_k \sim N(\mathbf{R}\mathbf{p}_k + \mathbf{t}, \mathbf{R}\mathbf{\Lambda}_k\mathbf{R}^T + \mathbf{\Xi}_k) \quad k = 1, \ldots, N \tag{3}$$

Collecting all N observations on $\mathcal{Q}$ into a single $3N$-dimensional vector $\mathbf{z}$, we have $\mathbf{z} \sim N(\boldsymbol{\mu}, \mathbf{\Gamma})$, where

$$\boldsymbol{\mu} = (\mathbf{I}_N \otimes \mathbf{R})\mathbf{p} + \mathbf{1}_N \otimes \mathbf{t} \tag{4}$$

is the mean of the observation vector and $\mathbf{\Gamma} = \mathrm{diag}(\mathbf{\Gamma}_1, \ldots, \mathbf{\Gamma}_N)$ is its $3N \times 3N$ block-diagonal covariance matrix, such that its k-th block is given by

$$\mathbf{\Gamma}_k = \mathbf{R}\mathbf{\Lambda}_k\mathbf{R}^T + \mathbf{\Xi}_k \ . \tag{5}$$

In this problem set-up the available noisy observation is $\mathbf{z}$. Thus, the log-likelihood function for estimating $\mathbf{R}, \mathbf{t}$ from the given observation is given by

$$L(\mathbf{R}, \mathbf{t}; \mathbf{z}, \mathbf{p}) = -\frac{1}{2}\log|\mathbf{\Gamma}| - \frac{1}{2}\{\mathbf{z} - \boldsymbol{\mu}\}^T \mathbf{\Gamma}^{-1} \{\mathbf{z} - \boldsymbol{\mu}\} \tag{6}$$

Maximizing (6) yields the MLE. Ignoring the $\frac{1}{2}\log|\mathbf{\Gamma}|$ term results in a weighted least squares problem. [2] provides a closed form solution to this constrained LS problem.

Similar formulation of the mathematical model is encountered in the problems of self-localization and navigation relative to a map or relative to known triangulation markers. In these cases $\mathbf{y}$ represents the markers‘ locations or the locations of keypoints on the map – which are known up to some measurement error. The registration task is then to estimate the rigid transformation, given a point cloud observation $\mathbf{z}$ and the known reference point cloud $\mathbf{p}$.

III. A General Form of the Cramer-Rao bound (CRB)

For any $\mathbf{R} \in SO(3)$, there exist (not necessarily unique) $\boldsymbol{\omega} \in \mathbb{R}^3, ||\boldsymbol{\omega}|| = 1$ and $\theta \in \mathbf{R}$ such that $\mathbf{R} = e^{\mathbf{M}(\boldsymbol{\omega})\theta}$, where $\mathbf{M}(\boldsymbol{\omega})$ is the skew symmetric matrix

$$\mathbf{M}(\boldsymbol{\omega}) = \begin{bmatrix} 0 & -\boldsymbol{\omega}_3 & \boldsymbol{\omega}_2 \\ \boldsymbol{\omega}_3 & 0 & -\boldsymbol{\omega}_1 \\ -\boldsymbol{\omega}_2 & \boldsymbol{\omega}_1 & 0 \end{bmatrix} \tag{7}$$

This representation implies that any rotation matrix can be represented as a rotation around a fixed axis $\boldsymbol{\omega}$ by an angle θ, where $\theta = \arccos(\frac{\mathrm{tr}(\mathbf{R})-1}{2})$ and

$$\boldsymbol{\omega} = \frac{1}{2\sin(\theta)} \begin{bmatrix} \mathbf{R}_{3,2} - \mathbf{R}_{2,3} \\ \mathbf{R}_{1,3} - \mathbf{R}_{3,1} \\ \mathbf{R}_{2,1} - \mathbf{R}_{1,2} \end{bmatrix} \tag{8}$$

In this paper we are interested in analyzing and lower-bounding the error in estimating the object rigid transformation. Therefore, in order to overcome the non-uniqueness of this representation, in the following it is assumed that $0 < \theta < \pi$. This implies that, in fact, we are looking to estimate $\boldsymbol{\nu} = \boldsymbol{\omega}\theta$ where $0 < ||\boldsymbol{\nu}|| < \pi$. Hence in the following we shall refer to $\mathbf{M}(\boldsymbol{\nu})$ as the skew symmetric matrix in the representation of $\mathbf{R}$ as $\mathbf{R} = e^{\mathbf{M}(\boldsymbol{\nu})}$.

Earlier derivations of the CRB on $SO(3)$, [22], are derived as special cases of general derivations of a CRB on Riemannian manifolds, *e.g.*, [23], [24], by considering $\mathbf{R}$ to be the parameter of interest such that it is an element of the Riemannian manifold $SO(3)$. In the current derivation, following

[25], we take advantage of the above simple representation of a rotation matrix $\mathbf{R} \in SO(3)$ by a 3-D vector $\boldsymbol{\nu}$ to provide a simple formulation of the derivation of the CRB matrix on the error covariance in estimating the parameters of a rigid transformation, along the lines of the standard derivation where the parameter vector is a vector in Euclidean space. This implementation takes into account the constraints on $\boldsymbol{\nu}$ as a representation of a rotation matrix in $SO(3)$ during the evaluation of the derivatives with respect to $\boldsymbol{\nu}$ required in evaluating the Fisher Information Matrix (FIM).

The parameter vector $\boldsymbol{\xi}$ representing some rigid transformation in $SE(3)$ is then given by a 6-D vector $\boldsymbol{\xi} = [\boldsymbol{\nu}^T, \mathbf{t}^T]^T$. The estimation error is $\boldsymbol{\xi} - \hat{\boldsymbol{\xi}}$.

Earlier works, [25], [22], apply the results of their derivations of the CRB on $SO(3)$ to the Wahba problem which is a special case of the model in (3) where the transformation is rotation, the disturbance model is isotropic white observation noise, and the points in $\mathcal{P}$ are deterministic and known. In this paper we extend these results and derive the CRB for the case where the transformation is rigid and the point cloud measurements are subject to more general noise models, aiming at modeling both the sampling discrepancies of the point clouds that result from their different sampling patterns, as well as the additive observation noise model, which is not necessarily white and isotropic. Thus, the considered noise model is *nonstationary* and is determined by the local geometry of the object at every point.

The general expression for the Fisher Information Matrix (FIM) of a real Gaussian process is given by, see *e.g.*, [26],

$$[\mathbf{J}(\boldsymbol{\xi})]_{k,\ell} = \frac{\partial \boldsymbol{\mu}^T}{\partial \boldsymbol{\xi}_k}\boldsymbol{\Gamma}^{-1}\frac{\partial \boldsymbol{\mu}}{\partial \boldsymbol{\xi}_\ell} + \frac{1}{2}\mathrm{tr}\left\{\boldsymbol{\Gamma}^{-1}\frac{\partial \boldsymbol{\Gamma}}{\partial \boldsymbol{\xi}_k}\boldsymbol{\Gamma}^{-1}\frac{\partial \boldsymbol{\Gamma}}{\partial \boldsymbol{\xi}_\ell}\right\}\ , \quad (9)$$

where $\boldsymbol{\xi}$ is the parameter vector, $\boldsymbol{\mu}$ is the mean of the observation vector, $\boldsymbol{\Gamma}$ is the observation vector covariance matrix, and $[\mathbf{J}(\boldsymbol{\xi})]_{k,\ell}$ denotes the (k,ℓ) entry of $\mathbf{J}$.

Since the observation vector $\mathbf{z} \sim N(\boldsymbol{\mu}, \boldsymbol{\Gamma})$ as defined in (3)-(5) we have

$$\frac{\partial \boldsymbol{\mu}}{\partial \boldsymbol{\nu}_\ell} = (\mathbf{I}_N \otimes \frac{\partial \mathbf{R}}{\partial \boldsymbol{\nu}_\ell})\mathbf{p} \quad (10)$$

Let $\mathbf{e}_\ell$ denote the 3-dimensional vector where its ℓ-th element is 1, while all others are zero. Next, following [25], [27] it can be shown that

$$\frac{\partial \mathbf{R}}{\partial \boldsymbol{\nu}_\ell} = \mathbf{M}(\boldsymbol{\gamma}_\ell)\mathbf{R} \quad (11)$$

where

$$\boldsymbol{\gamma}_\ell = \mathbf{e}_\ell + \frac{1-\cos(||\boldsymbol{\nu}||)}{||\boldsymbol{\nu}||^2}\boldsymbol{\nu}\times\mathbf{e}_\ell + \frac{||\boldsymbol{\nu}|| - \sin(||\boldsymbol{\nu}||)}{||\boldsymbol{\nu}||^3}\boldsymbol{\nu}\times(\boldsymbol{\nu}\times\mathbf{e}_\ell) \quad (12)$$

Hence,

$$\frac{\partial \boldsymbol{\mu}}{\partial \boldsymbol{\nu}_\ell} = (\mathbf{I}_N \otimes [\mathbf{M}(\boldsymbol{\gamma}_\ell)\mathbf{R}])\mathbf{p} \quad (13)$$

Clearly,

$$\frac{\partial \boldsymbol{\mu}}{\partial \mathbf{t}_\ell} = \mathbf{1}_N \otimes \mathbf{e}_\ell \quad (14)$$

The k-th block of $\frac{\partial \boldsymbol{\Gamma}}{\partial \boldsymbol{\nu}_\ell}$ is given by

$$\begin{aligned}\frac{\partial \boldsymbol{\Gamma}_k}{\partial \boldsymbol{\nu}_\ell} &= \frac{\partial \mathbf{R}}{\partial \boldsymbol{\nu}_\ell}\boldsymbol{\Lambda}_k\mathbf{R}^T + \mathbf{R}\boldsymbol{\Lambda}_k(\frac{\partial \mathbf{R}}{\partial \boldsymbol{\nu}_\ell})^T \\ &= \mathbf{M}(\boldsymbol{\nu}_\ell)\mathbf{R}\boldsymbol{\Lambda}_k\mathbf{R}^T - \mathbf{R}\boldsymbol{\Lambda}_k\mathbf{R}^T\mathbf{M}(\boldsymbol{\nu}_\ell) \quad (15)\end{aligned}$$

Also, $\frac{\partial \boldsymbol{\Gamma}}{\partial \mathbf{t}_\ell} = \mathbf{0}$. Substituting (10)-(15) into (9) and taking its inverse yields a closed form expression for the CRB on the error covariance of any unbiased estimator of the rigid transformation parameters when only a single observation is available and $\mathbf{p}$ is known. We thus conclude that the FIM is not block diagonal which implies that there is coupling between the estimation errors of all six problem parameters, and that the expression of the bound is independent of the magnitude of the unknown translation $\mathbf{t}$.

IV. Experimental Results

We next evaluate the CRB and the performance of prominent solutions to the problem of estimating $\mathbf{R}$ and $\mathbf{t}$ from the observed point cloud $\mathbf{z}$ and the *a-priori* known point cloud $\mathbf{p}$, by fitting $\mathbf{z}_k$ to $\mathbf{p}_k$, for every k, as posed in Section II. In the following, we first validate the derivation of the CRB by evaluating the performance of two classical algorithms: the MLE, (6), and the constrained LS solution [2] that often serve as the transformation estimation backbone in many point cloud registration algorithms, including both RANSAC and ICP. We compare the performance of the MLE and LS algorithms to the CRB. In this set-up the experimental data is generated by employing a standard isotropic additive noise model to a synthetic 3D-model [28]. Next, a practical application of the derived CRB is presented, where we derive a method for selecting the so called "inlier threshold" commonly used by registration algorithms to identify point correspondences under an assumed estimated transformation – thus providing a principled alternative to currently employed heuristics. The method is demonstrated using the KITTI odometry dataset [29].

A. CRB Validation

In this set of experiments it is assumed that $\{\mathbf{p}_k\}_{k=1}^N$ is known, while $\{\mathbf{z}_k\}_{k=1}^N$ is subject to both the markers placement noise and measurement noise as defined in (3).

Figure 1 depicts the MLE and LS estimation error variance, compared to the CRB. The estimation results are based on 1000 independent Monte-Carlo experiments, where in each experiment the obtained LS estimate is employed as initialization to the ML optimization procedure on SE(3). The experimental results indicate that in the investigated experiment both the LS and MLE are unbiased. It is further concluded that both the LS and ML estimators closely follow the CRB.

B. CRB Guided "Inlier Threshold" Selection

Many registration algorithms (*e.g.*, RANSAC [1], ICP [4], Super4PCS [30], TEASER++ [31]), involve a parameter that determines, under an assumed estimated transformation, which putative matches are considered as inliers, *i.e.*, let δ denote the inlier threshold, and $\widehat{T}$ an estimated transformation. The inlier

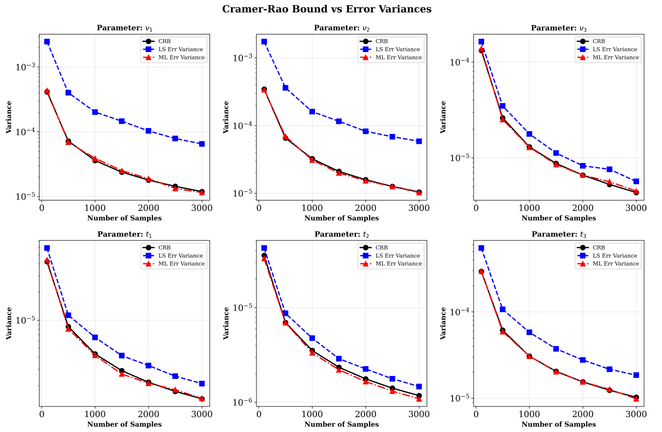

Fig. 1. Single observation and known reference points: Error Variance of ξ using the maximum likelihood estimator, constrained LS and the CRB.

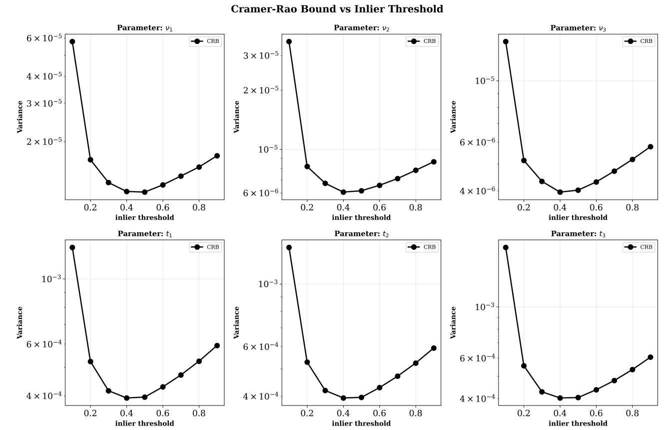

Fig. 2. CRB as function of the inlier threshold parameter. The effect of the tradeoff between the required accuracy of each inlier match and the resulting size of the inlier set is shown, while also showing that there is an optimal threshold value.

set, C, is then determined by $C = \{(\mathbf{p}_i, \mathbf{z}_i) | \left\| \widehat{T}(\mathbf{p}_i) - \mathbf{z}_i \right\| < \delta\}$. For example, this inlier set is used in RANSAC to determine the consensus size, or in ICP to select which of the putative matches are selected for estimating the transformation in the current iteration. Thus, the value of δ has great effect on the potential performance of these algorithms: By tightening the threshold, the effective standard deviation of the sampling noise is decreased, but then the set of inliers becomes smaller. On the other hand, increasing the threshold results in a larger sampling noise and a larger inlier set. It is *a-priori* unclear which effect will dominate, and it may depend on the problem parameters. Hence, this parameter is traditionally selected heuristically. As a consequence, analyzing the performance bound as a function of this threshold is of significant value in providing a principled method for the threshold selection.

In the following, a simple process for selecting the inlier threshold using the CRB, is proposed and demonstrated. The considered example is demonstrated on a scan from the KITTI odometry dataset [29]: We generate the set $\{\mathbf{p}_k\}_{k=1}^N$, by randomly sampling N points from the scan. To simulate a realistic scenario of obtaining the observed $\{\mathbf{y}_k\}_{k=1}^N$, we then downsample the original scan using voxel downsampling with $30cm$ resolution and find the nearest points to $\{\mathbf{p}_k\}_{k=1}^N$, yielding $\{\mathbf{y}_k\}_{k=1}^N$, where it is assumed that $\mathbf{y}_k \sim N(\mathbf{p}_k, \Lambda_k)$. Next, by randomly sampling N points from the *transformed* original scan, and finding the nearest neighbors to $\mathbf{R}\{\mathbf{y}_k\}_{k=1}^N + \mathbf{t}$, we obtain $\{\mathbf{z}_k\}_{k=1}^N$, such that $\mathbf{z}_k \sim N(\mathbf{R}\mathbf{p}_k + \mathbf{t}, \mathbf{R}\Lambda_k\mathbf{R}^T + \Xi_k)$.

This process simulates registration performed between a reference model, and a scanned model, including the commonly performed pre-processing steps: voxel downsampling, and random downsampling. Λ_k is assumed to be diagonal with standard deviation that is quarter of the voxel downsampling resolution, while Ξ_k is assumed to be diagonal with standard deviation that is half of the inlier threshold. At each inlier threshold, the set of inliers is found using the ground truth transformation, and the CRB is calculated for this set of inliers. Figure 2 depicts the resulting CRB as a function of the inlier threshold. In the figure, the tradeoff discussed earlier between the required accuracy of each inlier match and the resulting size of the inlier set is shown, while also showing that there is an optimal threshold value that can be selected based on the problem statistics.

V. Conclusions

This paper provides a simple derivation of a closed form, easy to compute expression for the Cramer-Rao lower bound on the error covariance of any unbiased estimator of the rigid transformation parameters in the problem of point cloud to known landmarks registration. This type of formulation is encountered in problems of self-localization and navigation relative to a map or relative to known markers, as well as in medical imaging where pre-installed markers are employed. Furthermore, while existing analyses of the estimation performance of these algorithms in the presence of noise and model mismatches are restricted to naive models of additive isotropic measurement noise, we derive the bound subject to more general noise models, aiming at modeling both the sampling discrepancies of the point clouds , as well as the point-wise geometry dependent statistics of the noise.

Since most registration algorithms, learning-based, or classic, rely on RANSAC for initial alignment and on ICP for refined final alignment, analyzing their performance relative to the CRB is widely applicable and is valid for almost any existing registration algorithm. Since in both ICP and RANSAC the transformation estimation backbone usually employs the closed form LS solution [2], we experimentally validate both the correctness of the CRB derivation and the LSE efficiency relative to the CRB in the case where outliers are mitigated. Moreover, we derive a CRB-based selection rule for rigorously establishing the “inlier threshold” which defines which putative matches are considered inliers - thus providing a principled alternative to currently employed heuristics.

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